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ITA0329 – Mobile Computing: Assignment 3 (CO3)

Topic coverage: Introduction to wireless networks, Mobile IP, issues in Mobile IP, cellular vs ad hoc wireless networks, applications, routing protocols (proactive and reactive)

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Q1 – Wireless Network Architecture and Mobility Analysis (6 Marks)

Part A – Introduction to Wireless Networks

1.1 Shared Group Work

1.1.1 Characteristics of wireless networks

A wireless network connects devices using radio-frequency (RF), microwave or infrared signals instead of physical cables. The main characteristics that distinguish it from a wired network are:

- **Shared, broadcast medium:** the air is shared by every transmitter in range, so access must be coordinated (CSMA/CA in Wi-Fi, TDMA/FDMA/OFDMA in cellular).
- **Limited and variable bandwidth:** capacity is a few Mbps to a few hundred Mbps per cell and depends on distance, interference and modulation, whereas a wired link offers a stable Gbps-class channel.

- **High and bursty error rate:** bit error rate (BER) is 10^-3 to 10^-5 on wireless links versus 10^-9 or better on fibre, because of fading, multipath, shadowing and interference.
- **Mobility:** the user can move, which changes signal quality, the serving cell and sometimes the IP network.
- **Limited power:** battery-powered devices must save energy, so they sleep, reduce transmit power and cannot stay on continuously.
- **Security exposure:** anyone within range can listen, so authentication and encryption are mandatory.
- **Asymmetry and hidden/exposed terminals:** two nodes may both hear an access point but not each other (hidden terminal), causing collisions.

1.1.2 Components of a wireless network

Component	Description	Example
Mobile Node (MN)	End-user device that moves and communicates wirelessly	Smartphone, laptop, tablet, IoT sensor
Base Station (BS) / Access Point (AP)	Fixed radio transceiver that provides the wireless access link and connects MNs to the wired network	Cellular tower (eNodeB/gNodeB), Wi-Fi AP
Wireless link	The radio channel between MN and BS (uplink and downlink)	4G/5G air interface, IEEE 802.11
Access/backhaul network	Wired or microwave network linking BSs to the core	Fibre ring, microwave backhaul

Component	Description	Example
Gateway / Router	Connects the wireless network to the Internet or other networks	Edge router, PDN gateway
Mobility-management entities	Keep track of the location of MNs	HLR/VLR, MME, Home Agent/Foreign Agent
Servers / Correspondent nodes	The other end of the communication	Web server, remote user

1.1.3 Developing a suitable architecture

The group selected an **infrastructure-based wireless architecture** with three coverage areas (cells/WLAN cells) connected to a common wired backbone. It was chosen because it clearly shows the interaction between mobile nodes, access points, the backbone and a fixed server, and it allows a mobility (handoff) scenario to be created between two cells.

The design has these layers:

1. **Access layer:** mobile nodes MN1, MN2, MN3 attach to access points AP1, AP2, AP3 over wireless links.
2. **Distribution layer:** the access points connect to an aggregation switch/router over wired links.
3. **Core/gateway layer:** a gateway router connects the network to the Internet and to the application server (correspondent node).

Figure 1.1 shows this architecture (diagram is inserted below the text of Q1).

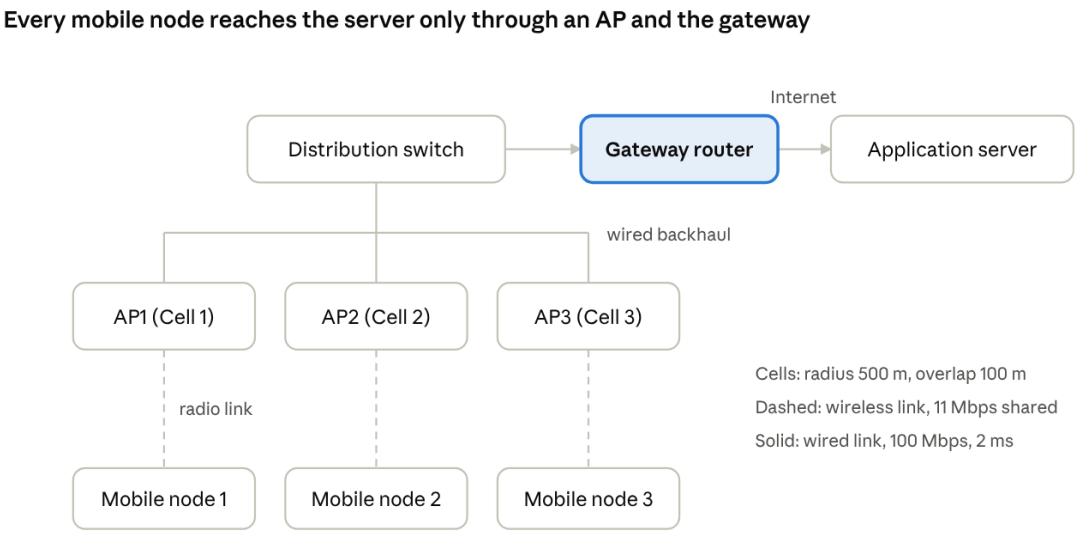


Figure 1.1 · wireless network architecture, 3 cells

Figure 1.1 – Infrastructure-based wireless network architecture used for the analysis.

1.1.4 Bandwidth, connectivity and mobility constraints

Constraint	Cause	Consequence for the application
Bandwidth	Limited spectrum, sharing among all users in a cell, modulation fallback at the cell edge	Throughput per user falls as users increase; video may degrade
Connectivity	Fading, shadowing, interference, coverage holes,	Intermittent disconnection, retransmissions, higher delay

Constraint	Cause	Consequence for the application
	obstacles	
Mobility	Movement across cells and networks	Handoff, address change, packet loss, session interruption
Energy	Battery life	Reduced transmit power and sleep periods increase latency
Security	Open radio medium	Need for authentication, encryption, key management

1.1.5 Communication between mobile users and infrastructure

A mobile node communicates in the following way: (1) the MN scans and associates with the AP having the strongest signal; (2) the MN authenticates and receives an IP address (DHCP) or a care-of address; (3) uplink frames are sent to the AP using the medium access protocol; (4) the AP forwards them over the wired distribution network to the gateway; (5) the gateway routes them over the Internet to the correspondent node; (6) the reply follows the reverse path, and the AP transmits on the downlink.

1.1.6 Small simulated communication scenario

A small scenario was simulated to test the communication flow.

Setup: 3 APs placed 900 m apart, each with a 500 m radius; one MN walks/drives from AP1 to AP3 along a straight road; the MN sends a constant-bit-rate (CBR) UDP stream of 50 packets/s (160 bytes each, i.e. a voice-like stream) to a fixed server; the wired backbone has 100 Mbps

links with 2 ms delay; the wireless link rate is 11 Mbps (802.11b-like) with a 20 ms mean one-way access delay; handoff interruption time is 300 ms.

Observed flow (test of communication):

Time (s)	Event	Result
0	MN associated with AP1, stream starts	Packets delivered, one-way delay about 40 ms
40	Signal from AP1 falls below threshold, MN scans	Delay rises to about 55 ms
41	MN reassociates with AP2	About 15 packets lost during the 300 ms handoff
42	Stream resumes through AP2	Delay returns to about 40 ms
96	Second handoff AP2 to AP3	Again about 15 packets lost
120	Stream ends	Total sent 6000, delivered 5970, delivery ratio 99.5%

The test confirmed the end-to-end path MN → AP → switch → gateway → server and showed that every handoff costs a short burst of packet loss.

1.2 Student A – Network Architecture Analysis

1.2.1 Wireless network architecture representation

The architecture is shown in **Figure 1.1**. The wireless part uses a star topology around each AP (cell), while the wired part is a tree/star topology connecting the APs to the gateway.

1.2.2 Major components and their roles

- **Mobile nodes (MN1–MN3):** generate and consume traffic, measure signal strength, initiate association and handoff.
- **Access points (AP1–AP3):** provide the radio interface, manage association, perform medium access control, translate between 802.11 frames and wired Ethernet frames.
- **Distribution switch:** forwards frames between APs and the gateway, isolates traffic between cells.
- **Gateway router:** performs IP routing, NAT, firewall and connection to the Internet.
- **DHCP/authentication server:** assigns addresses, authenticates users (RADIUS/802.1X).
- **Application server (correspondent node):** provides the service used by the mobile users.

1.2.3 Wireless communication characteristics

Path loss increases with distance d as 1/d^n where n is 2 in free space and 3 to 4 in urban areas. Received power falls by about 20n dB per decade of distance, so the data rate must be reduced at the cell edge (adaptive modulation such as 64-QAM → QPSK). The channel suffers from multipath fading, which causes bursty errors; it is half-duplex in Wi-Fi (one direction at a time) and uses collision avoidance rather than collision detection, because a node cannot hear while it transmits.

1.2.4 Effect of limited bandwidth and wireless connectivity

The per-user throughput in a cell is approximately:

Throughput per user = (Cell capacity × MAC efficiency) / N

For an 11 Mbps cell with MAC efficiency about 55% the usable capacity is about 6 Mbps. Table 1.1 and Graph 1.1 show how the per-user throughput falls with the number of active users.

Table 1.1 – Per-user throughput versus number of users (usable cell capacity = 6 Mbps)

Active users N	Throughput per user (Mbps)	Delay factor (relative)
1	6.00	1.0
5	1.20	1.4
10	0.60	2.1
20	0.30	3.8
40	0.15	7.5

With poor connectivity (low SNR) the link layer retransmits frames, so effective throughput falls and delay rises; a link with a 20% frame error rate needs on average $1/(1 - 0.2) = 1.25$ transmissions per frame, which reduces goodput by 20% and adds about 25% to mean access delay. For real-time services such as voice, packets arriving later than 150 ms are useless, so the delay increase is as damaging as loss.

1.2.5 How mobile nodes communicate through the wireless infrastructure

Mobile nodes never communicate directly with each other in infrastructure mode; all traffic goes through the AP/base station, even between two nodes in the same cell. This centralised

arrangement allows the infrastructure to control access, security and handoff, but it means the AP is a single point of failure and a capacity bottleneck. Each frame carries the BSSID of the AP, the source and the destination MAC addresses; the AP rewrites the header when forwarding between the wireless and wired sides.

1.3 Student B – Mobility and Performance Analysis

1.3.1 Representative mobility scenarios

Scenario	Speed	Typical situation	Expected handoffs
S1 – Stationary	0 km/h	Student at desk	None
S2 – Pedestrian	5 km/h	Walking on campus	Rare
S3 – Cyclist/slow vehicle	30 km/h	Campus road	Occasional
S4 – City vehicle	60 km/h	Ambulance in city	Frequent
S5 – Highway	120 km/h	Highway/train	Very frequent

The handoff rate depends on the speed v and on the cell radius R. For a straight crossing the mean path through a circular cell is about πR/2, so

Handoffs per minute = (v × 60) / (πR / 2)

With R = 500 m, πR/2 ≈ 785 m.

1.3.2 Handoff and connectivity interruption

A handoff has four phases: **(1) measurement/detection** of falling signal, **(2) scanning and target selection**, **(3) authentication and reassociation** with the new AP, and **(4) resumption of traffic**. The interruption (the time the MN cannot send or receive) is the sum of these phases:

T_interrupt = T_detect + T_scan + T_auth + T_assoc (+ T_registration if the IP network changes)

Typical values: detection 20 ms, active scan 100 ms, authentication 50 ms, association 30 ms → about 200 ms; with Mobile IP registration of about 100 ms the total is about 300 ms. A hard handoff (break-before-make) has this interruption; a soft handoff (make-before-break, as in CDMA) reduces it to nearly zero.

1.3.3 Packet loss and transmission delay

For a CBR source of rate λ packets/s and an interruption time T_interrupt, the number of packets lost per handoff is about **λ × T_interrupt**. With λ = 50 pkt/s and T = 0.3 s this is 15 packets per handoff. The loss ratio is therefore:

Loss % = (handoffs per second × λ × T_interrupt / λ) × 100 = handoffs per second × T_interrupt × 100

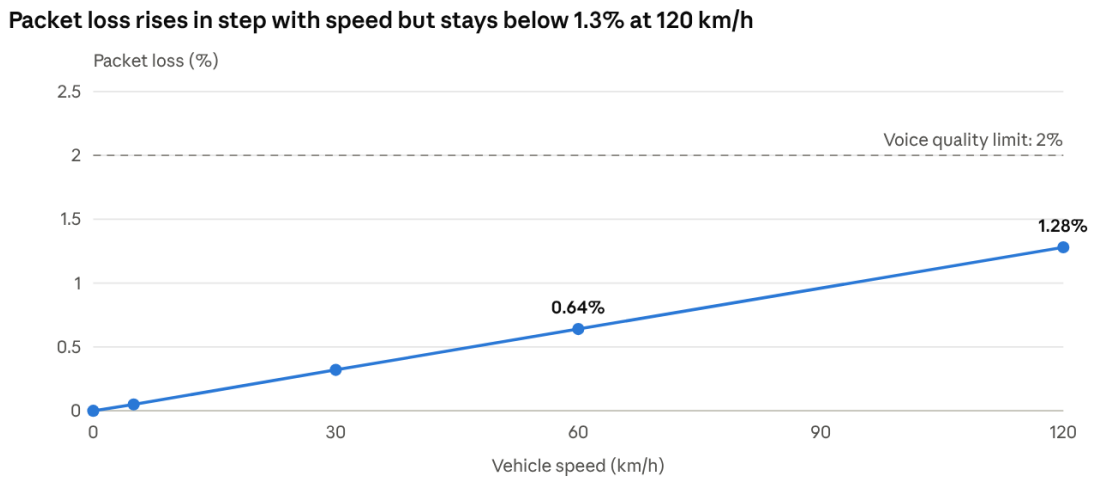
This equals 0.5% per (handoff/min) for T = 0.3 s. The delay also increases because packets are queued during handoff and because the signal at the cell edge forces retransmissions.

1.3.4 Effect of changing network conditions and comparison under different mobility

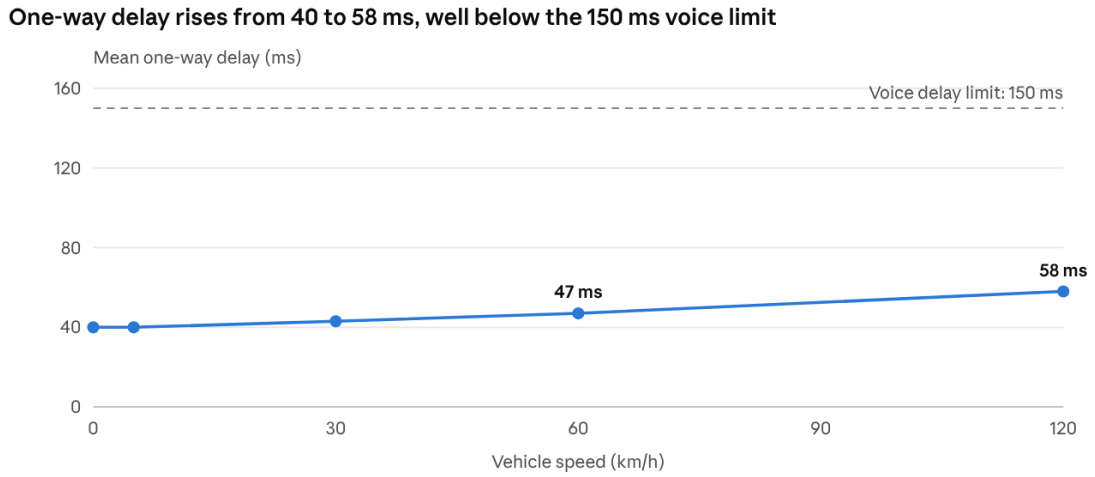
Table 1.2 – Effect of mobility on communication (R = 500 m, T_interrupt = 0.3 s, 50 pkt/s)

Scenario	Speed (km/h)	Handoffs per min	Packet loss (%)	Mean one-way delay (ms)	Connection interrupted time per min (s)
S1 Stationary	0	0	0.00	40	0.0
S2 Pedestrian	5	0.11	0.05	40	0.03
S3 Slow vehicle	30	0.64	0.32	43	0.19
S4 City vehicle	60	1.27	0.64	47	0.38
S5 Highway	120	2.55	1.28	58	0.77

Observations: (i) the number of handoffs increases linearly with speed; (ii) packet loss is proportional to handoff frequency; (iii) delay rises faster than linearly at high speed because the MN spends more time at the weak signal edge and because retransmissions and re-registration queue the traffic. When changing conditions occur, such as a sudden increase in the number of users or interference, the AP lowers its modulation rate, so the same packet occupies the channel for longer and the delay of all users increases. **Graph 1.1** and **Graph 1.2** summarise these results.



Graph 1.1 · analytical model, Table 1.2 values



Graph 1.2 · analytical model, Table 1.2 values

Graph 1.1 (loss versus speed) and **Graph 1.2** (delay versus speed) are drawn from the values of Table 1.2.

Part B – Wireless Mobility Scenario

1.4 Shared Group Work – the scenario

A mobile user (a student with a smartphone making a VoIP call to a fixed server) moves along a campus road through three adjacent coverage areas: Cell 1 (AP1), Cell 2 (AP2) and Cell 3 (AP3). Each cell has a 500 m radius and adjacent cells overlap by about 100 m. The user starts in Cell 1 at the left end, moves at constant speed to the right, and the call must continue until the user reaches the end of Cell 3. **Figure 1.2** shows the movement path and the overlap regions.

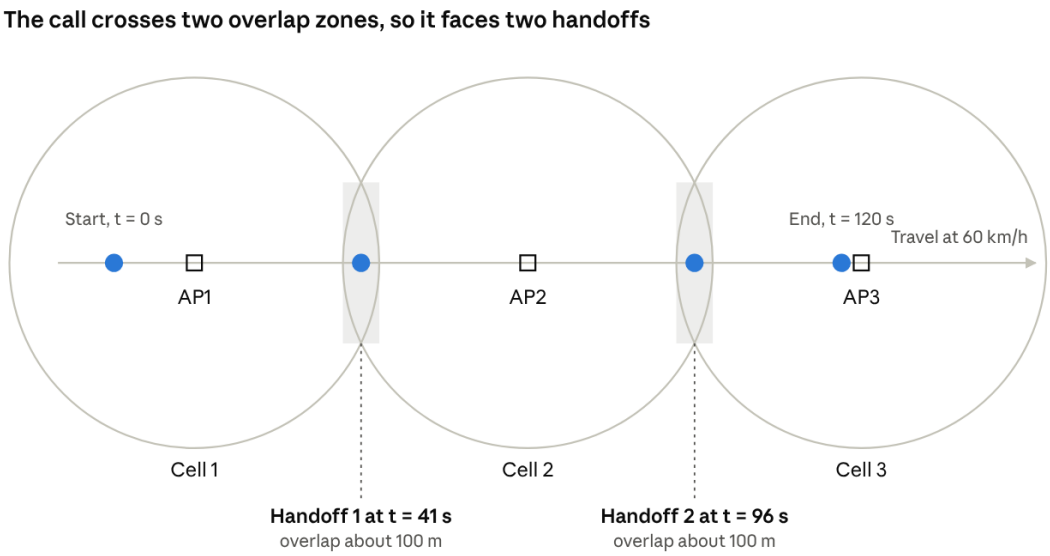


Figure 1.2 · mobility scenario across three cells

Figure 1.2 – Movement of the mobile user through three overlapping cells (distances to scale: cell spacing 900 m, radius 500 m).

1.5 Student A – Analytical Investigation

1.5.1 Successful and interrupted communication cases

Case	Condition	Outcome
Successful communication	Signal ≥ threshold (e.g. −80 dBm) and packet delay ≤ 150 ms and loss ≤ 2%	Call quality acceptable
Degraded communication	Signal between −80 dBm and −90 dBm, high retransmissions	Delay and jitter rise; still usable
Interrupted communication	Handoff in progress, or signal < −90 dBm with no alternative AP	No packets delivered for T_interrupt
Failed communication	New AP rejects association (overload) or registration fails	Call dropped

1.5.2 Sequence of mobility events

1. MN attached to AP1, signal strong, traffic flows.
2. MN moves toward the overlap area; RSSI from AP1 falls and RSSI from AP2 rises.
3. RSSI(AP1) drops below the handoff threshold; MN starts scanning.
4. MN selects AP2 (RSSI(AP2) > RSSI(AP1) + hysteresis margin of 5 dB).
5. MN authenticates and reassociates with AP2; traffic is interrupted.
6. Registration/location update completes; traffic resumes through AP2.
7. Steps 2–6 repeat for the AP2 → AP3 transition.
8. MN reaches the end of Cell 3; the call finishes.

1.5.3 Delay and packet-delivery behaviour (recorded)

Table 1.3 – Recorded values for one 120 s run at 60 km/h (16.7 m/s), starting 220 m before AP1

Time window (s)	Serving AP	Packets sent	Packets delivered	Mean delay (ms)
0–35	AP1	1750	1750	40
35–42 (handoff 1 at 41 s)	AP1 → AP2	350	335	55
42–95	AP2	2650	2650	40
95–102 (handoff 2 at 96 s)	AP2 → AP3	350	335	54
102–120	AP3	900	900	40
Total		**6000**	**5970**	**42 (overall)**

Delivery ratio = 5970 / 6000 = 99.5%.

1.5.4 Increasing mobility conditions

When the speed is doubled from 30 to 60 km/h (the recorded run) the road is crossed in half the time, handoffs come twice as often, the overlap time is halved (about 6 s at 60 km/h versus 12 s at 30 km/h), and the margin for completing the handoff before the old link dies is smaller. At 120 km/h the overlap lasts only about 3 s, so with a 3 s scan + handoff procedure the connection may drop before the new AP is ready. Hence the required handoff time must be less than overlap distance / speed:

T_handoff < D_overlap / v

For D_overlap = 100 m: at 30 km/h limit = 12 s; at 60 km/h = 6 s; at 120 km/h = 3 s.

1.5.5 Analytical evidence

The evidence consists of Table 1.2, Table 1.3, Graph 1.1 (handoffs and loss versus speed) and Graph 1.2 (delay versus speed): the loss ratio grows linearly with speed, the delay grows non-linearly, and the call remains acceptable (loss < 2%, delay < 150 ms) for speeds up to highway speed provided T_interrupt stays below the overlap time.

1.6 Student B – Mobility Logic

1.6.1 Mobility decision sequence

The decision logic of the mobile node is expressed using the standard pseudocode notation of the course:

```
ALGORITHM  DecideAttachment(MN, CurrentAP, NeighborAPs)
INPUT:
  MN          - mobile node
  CurrentAP    - AP currently serving MN
  NeighborAPs  - APs detected by scanning
OUTPUT:
  AttachmentDecision
BEGIN
  RSSI_cur <- SignalStrength(CurrentAP)
  IF RSSI_cur >= HANDOFF_THRESHOLD  THEN
    RETURN STAY_ATTACHED
  END IF
```

```
  BestAP <- NULL
  BestRSSI <- RSSI_cur
  FOR each AP in NeighborAPs DO
    IF SignalStrength(AP) > BestRSSI + HYSTERESIS THEN
      BestAP <- AP
      BestRSSI <- SignalStrength(AP)
    END IF
  END FOR
  IF BestAP = NULL THEN
    IF RSSI_cur < MIN_SIGNAL THEN
      RETURN CONNECTION_LOST
    ELSE
      RETURN STAY_ATTACHED
    END IF
  ELSE
    RETURN HANDOFF_TO(BestAP)
  END IF
END
```

1.6.2 When does a mobile node change its point of attachment?

A node changes its attachment when **all** of the following hold: the current signal is below the handoff threshold; a neighbouring AP is stronger by at least the hysteresis margin (this prevents the "ping-pong" effect where the node oscillates between two APs at the cell boundary); the target AP has free capacity; and authentication with the target succeeds. Without hysteresis, a

node at the boundary could perform many unnecessary handoffs per minute, each causing a loss burst.

1.6.3 Communication conditions affected

During a handoff the following are affected: **(a)** packet delivery (loss during T_interrupt), **(b)** delay and jitter (queueing before/after), **(c)** ongoing TCP connections (timeouts cause window reduction), **(d)** the IP address if the new AP belongs to another subnet (the session breaks unless Mobile IP is used), **(e)** security context (re-authentication needed) and **(f)** QoS reservations (must be re-established).

1.6.4 Termination / completion condition for the scenario

The scenario terminates successfully when the MN reaches the end of the coverage of Cell 3 and the application session is closed normally, or terminates with failure when (i) signal < MIN_SIGNAL and no neighbour AP exists (coverage hole), (ii) registration/association fails and cannot be restored, or (iii) the application session timeout expires.

1.6.5 Repeated mobility and communication efficiency

If the node performs k handoffs per session, each costing T_interrupt seconds of lost communication, the effective efficiency is

Efficiency = 1 − (k × T_interrupt) / T_session

For a 120 s session with k = 2 and T = 0.3 s the efficiency is 1 − 0.6/120 = 99.5%. If the node ping-pongs 20 times, efficiency falls to 95%, and the signalling load of 20 registrations also loads the network. Hence hysteresis, handoff-prediction and fast handoff are used to maintain efficiency.

1.7 Individual verification (Q1)

- *Student A should be able to explain the mobility scenario:* the node moves across overlapping cells, hands off when RSSI falls below the threshold by the hysteresis margin, loses about $\lambda \times T$ packets per handoff, and resumes.
- *Student B should be able to explain the architecture:* MN → AP → distribution switch → gateway → server, with APs providing radio access and the gateway providing Internet connectivity.

Q2 – Mobile IP and Mobility Management

2.1 Shared Group Work

2.1.1 Mobile IP architecture and terminology

An IP address does two jobs: it identifies the host and it tells routers where the host is (its subnet). When a node leaves its subnet, packets addressed to the old address keep going to the old subnet, so TCP connections and applications break. Mobile IP (IPv4: RFC 5944) solves this by giving the mobile node a permanent **home address** for identification and a temporary **care-of address (CoA)** for location, and by keeping a **binding** between the two at a router on the home network.

Term	Meaning
Mobile Node (MN)	Host or router that changes its point of attachment and keeps its home address

Term	Meaning
Home network / home address	Network where the MN's permanent address belongs / the permanent address, e.g. 10.1.1.25
Home Agent (HA)	Router on the home network that keeps the binding (home address to CoA) and tunnels packets to the MN
Foreign network	Any network visited by the MN other than its home network
Foreign Agent (FA)	Router on the foreign network that advertises itself, relays registration and delivers decapsulated packets to the MN
Care-of Address (CoA)	Address valid in the foreign network where tunnels end: either the FA's address (FA-CoA) or an address obtained by the MN itself (co-located CoA)
Correspondent Node (CN)	The host communicating with the MN; it may be fixed or mobile
Binding	Association home address, CoA and lifetime stored in the HA binding table
Tunnel	IP-in-IP encapsulation from HA to the CoA (outer header protocol number 4)

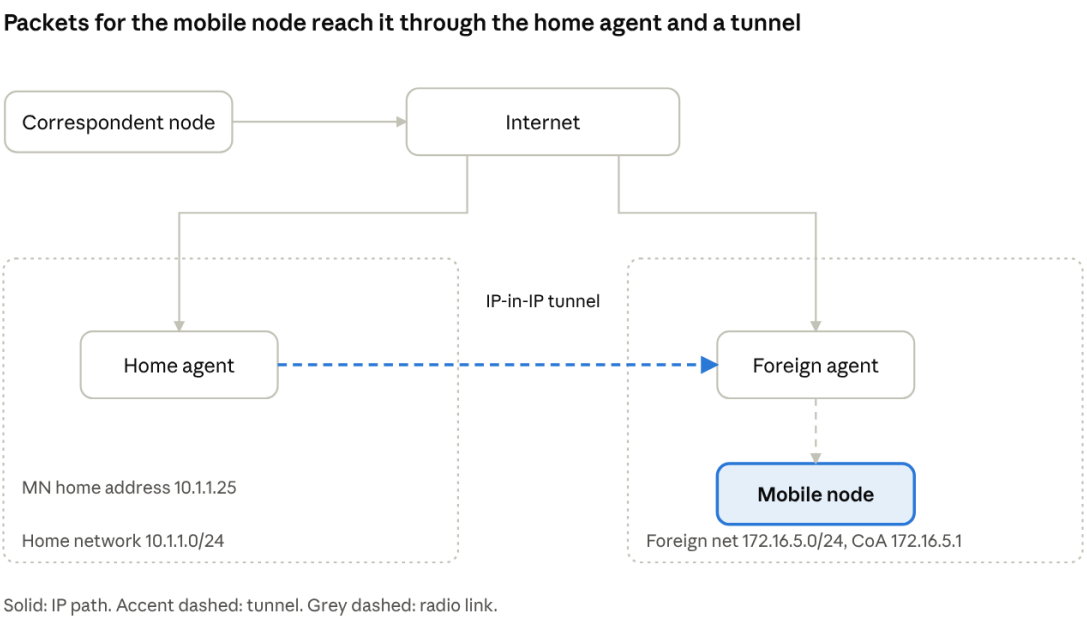


Figure 2.1 · Mobile IP architecture

Figure 2.1 – Mobile IP architecture: the CN sends to the home address, the HA intercepts and tunnels to the FA, and the FA delivers to the MN over the radio link.

2.1.2 Agent discovery

Mobility agents announce themselves by **agent advertisements**, ICMP router advertisements extended with a mobility-agent extension. They are sent periodically (for example once per second) and contain the agent's address, the CoAs offered, flag bits H (home agent) and F (foreign agent) and a registration lifetime. If the MN does not want to wait it sends an **agent solicitation**.

The MN detects that it has moved when (a) the advertisement lifetime of its current agent expires, or (b) the network prefix of the advertised agent differs from the previous one. If the advertisement comes from its own home agent, the MN is at home and deregisters; otherwise it is in a foreign network and must register.

```
ALGORITHM MobileIPAgentDiscovery(MN)
INPUT:
  MN - mobile node
OUTPUT:
  Detected mobility agent or AGENT_NOT_FOUND
BEGIN
  LISTEN for AgentAdvertisement
  IF Advertisement received THEN
    Extract NetworkAddress
    Extract AgentType
    IF AgentType = HOME_AGENT THEN
      Set HomeAgentAddress
    ELSE IF AgentType = FOREIGN_AGENT THEN
      Set ForeignAgentAddress
    END IF
    RETURN AgentType, NetworkAddress
  ELSE
    SEND AgentSolicitation
    WAIT for Advertisement
    IF Advertisement received THEN
      RETURN detected agent information
    ELSE
      RETURN AGENT_NOT_FOUND
```

```
  END IF
 END IF
END
```

2.1.3 Registration

Registration tells the home agent where the MN is. The **registration request** carries the home address, the home agent address, the CoA, the requested lifetime, a 64-bit identification field (timestamp or nonce, which defeats replay attacks) and the mobile-home authentication extension (a keyed hash computed with the secret shared by MN and HA). The **registration reply** carries a code (0 means accepted, the 64 to 88 range means denied by the FA and 128 to 139 means denied by the HA) and the granted lifetime. If the lifetime expires without renewal, the binding is deleted.

```
ALGORITHM RegisterMobileNode(MN, HA, CoA)
INPUT:
 MN - mobile node
 HA - home agent
 CoA - current care-of address
OUTPUT:
 Registration result
BEGIN
 Create RegistrationRequest
 Request.HomeAddress  <- MN.HomeAddress
 Request.CareOfAddress <- CoA
 Request.Lifetime     <- RequestedLifetime
```

```
  Request.Identifier    <- GenerateRegistrationIdentifier()
  Authenticate Request
  SEND Request TO HA
  WAIT FOR RegistrationReply
  IF Reply.Status = ACCEPTED THEN
    Binding[MN.HomeAddress] <- CoA
    RegistrationExpiry <- CurrentTime + Reply.Lifetime
    RETURN REGISTRATION_SUCCESS
  ELSE
    RETURN REGISTRATION_FAILURE
  END IF
END
```

2.1.3a Location update

Every time the MN attaches to a new network it computes a new CoA and repeats the registration, which updates the binding at the HA (location updating). The CN is never informed in basic Mobile IPv4; this keeps the CN unchanged, but it is the origin of triangular routing described in Student B's analysis.

2.1.4 Packet delivery and tunnelling

1. The CN sends a packet with source = CN and destination = MN's **home address**; ordinary IP routing carries it to the home network.
2. The HA intercepts it (proxy ARP on the home network) and finds the binding for the home address.

3. The HA **encapsulates** the packet: a new outer IP header with source = HA, destination = CoA is added in front of the original packet (IP-in-IP).
4. The tunnel delivers the packet to the CoA (the FA).
5. The FA **decapsulates** it and delivers the original packet to the MN over the radio link.
6. The MN replies with source = home address and destination = CN; the reply travels directly to the CN (basic Mobile IP), or through a reverse tunnel when ingress filtering is used.

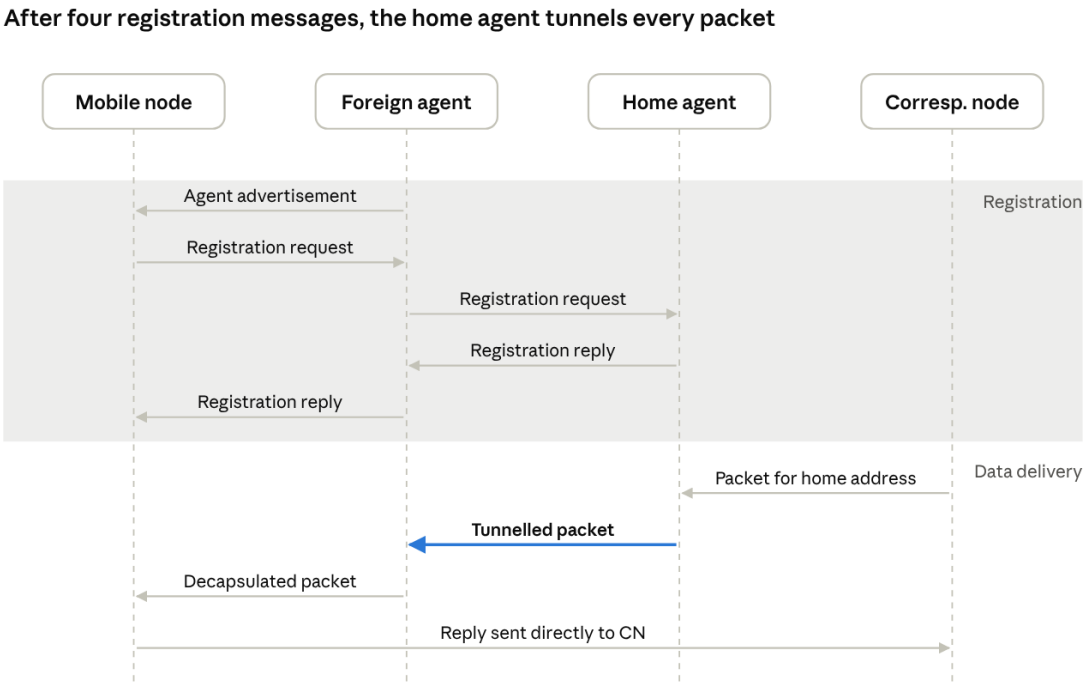


Figure 2.2 · registration and packet-flow sequence

Figure 2.2 – Registration and packet-flow sequence. The shaded band is the registration phase; below it, data delivery.

ALGORITHM TunnelPacketToMobileNode(Packet, DestinationHomeAddress)

INPUT:

```
  Packet              - original IP packet
  DestinationHomeAddress - mobile node home address
OUTPUT:
  Packet delivered through tunnel or DROP
BEGIN
  IF Binding[DestinationHomeAddress] does not exist THEN
    DROP Packet
    RETURN DELIVERY_FAILURE
  END IF
  CoA <- Binding[DestinationHomeAddress]
  EncapsulatedPacket <- ENCAPSULATE(Packet, CoA)
  FORWARD EncapsulatedPacket TO CoA
  IF TunnelDeliverySuccessful THEN
    Decapsulate at tunnel endpoint
    DELIVER original Packet TO MobileNode
    RETURN DELIVERY_SUCCESS
  ELSE
    RETURN DELIVERY_FAILURE
  END IF
END
```

2.1.5 Worked example (small simulation by hand)

Setup: home network 10.1.1.0/24 with HA 10.1.1.1; MN home address 10.1.1.25; foreign network 172.16.5.0/24 with FA 172.16.5.1 (FA-CoA); CN 192.168.9.7. Delays: radio hop MN-

FA 10 ms, FA-HA 50 ms one way, CN-HA 60 ms, processing 10 ms per registration message at each agent, agent advertisement every 1 s.

Step	Node	Event	Decision / action	State after
1	MN	Moves into 172.16.5.0/24 and hears advertisement from 172.16.5.1	Prefix differs from home, so MN is away	Location = FOREIGN
2	MN	Sends registration request (home 10.1.1.25, CoA 172.16.5.1, lifetime 300 s) to FA	Authentication extension added	Registration = PENDING
3	FA	Receives request	Records pending visitor entry, relays to HA	Visitor list updated
4	HA	Receives request	Checks keyed hash and identification, accepts	Binding 10.1.1.25 to 172.16.5.1, lifetime 300 s
5	HA, FA, MN	Reply travels HA, FA, MN	Code 0 (accepted)	Registration = REGISTERED
6	CN	Sends packet 192.168.9.7 to 10.1.1.25	Routed to home network	Packet arrives at HA

Step	Node	Event	Decision / action	State after
7	HA	Intercepts packet	Outer header 10.1.1.1 to 172.16.5.1 added (20 bytes)	Tunnelled
8	FA	Receives tunnelled packet	Removes outer header, delivers to 10.1.1.25	Delivered
9	MN	Replies 10.1.1.25 to 192.168.9.7	Sent directly via FA	Reply delivered

Registration delay = 2 x (10 + 50) ms of transmission + 4 x 10 ms processing (FA and HA on request and reply paths) = **160 ms**, plus the mean 500 ms wait for the next advertisement, giving about **660 ms** of interruption. A CBR stream at 50 packets/s loses about 33 packets, which the TCP or the codec must absorb.

2.2 Student A – Mobile IP Operation

2.2.1 Role of the Mobile Node

The MN keeps its permanent home address and uses it as the source address of all its packets. It listens to agent advertisements, decides whether it is at home or away, obtains a CoA, sends registration requests (before the previous lifetime expires), authenticates itself to the HA with the shared key, deregisters when it returns home, and, if a reverse tunnel is required, encapsulates its outgoing packets to the HA.

2.2.2 Role of the Home Agent

The HA keeps the binding table, authenticates and accepts or rejects registrations, intercepts packets addressed to home addresses of absent MNs (using proxy ARP and gratuitous ARP on the home link), encapsulates and tunnels them to the CoA, decapsulates reverse-tunnelled packets, and deletes bindings when the lifetime expires or when the MN deregisters.

2.2.3 Role of the Foreign Agent

The FA advertises its services, keeps a visitor list, relays registration requests to the HA and replies to the MN, acts as the tunnel end point (decapsulation), delivers packets to the MN using its home address at the link layer, and may act as the default router for the visiting MN. With a co-located CoA the FA is not needed.

2.2.4 Role of the Correspondent Node

The CN is a normal IP host. In basic Mobile IP it needs no modification: it sends to the MN's home address and does not know that the MN has moved. In route-optimisation schemes it keeps its own binding cache and sends directly to the CoA.

2.2.5 Registration and location updating

Registration is a four-message exchange (MN to FA, FA to HA, HA to FA, FA to MN) protected by the authentication extension. It is repeated (a) at every change of network, (b) before the lifetime expires (periodic refresh, for example every 150 s for a 300 s lifetime), and (c) with lifetime 0 for deregistration when the MN returns home. The HA binding table is the only location database; its freshness defines how well packets are delivered.

2.2.6 Packet delivery and tunnelling process

Forward path: CN, Internet, HA (encapsulate), tunnel, FA (decapsulate), MN. Reverse path: MN, FA, CN (direct) or MN, FA, HA (reverse tunnel), CN. Encapsulation adds a new 20-byte IP header and reduces the usable payload (MTU) by 20 bytes, which may cause fragmentation.

CoA type	Tunnel end point	Advantages	Disadvantages
Foreign-agent CoA	FA	One address shared by many MNs; saves IPv4 addresses; FA does the decapsulation	Needs FA infrastructure; FA is a bottleneck
Co-located CoA	MN itself	Works without FA (any DHCP network); decapsulation at MN	Uses one address per MN; MN does more work; radio carries the 20-byte overhead

2.3 Student B – Mobile IP Challenge Analysis

2.3.1 Identified challenges

Challenge	Cause	Effect	Remedy
Handoff / registration delay	Waiting for advertisement plus round trip to a distant HA	Hundreds of ms of interruption	Regional registration, fast handover, link-layer triggers

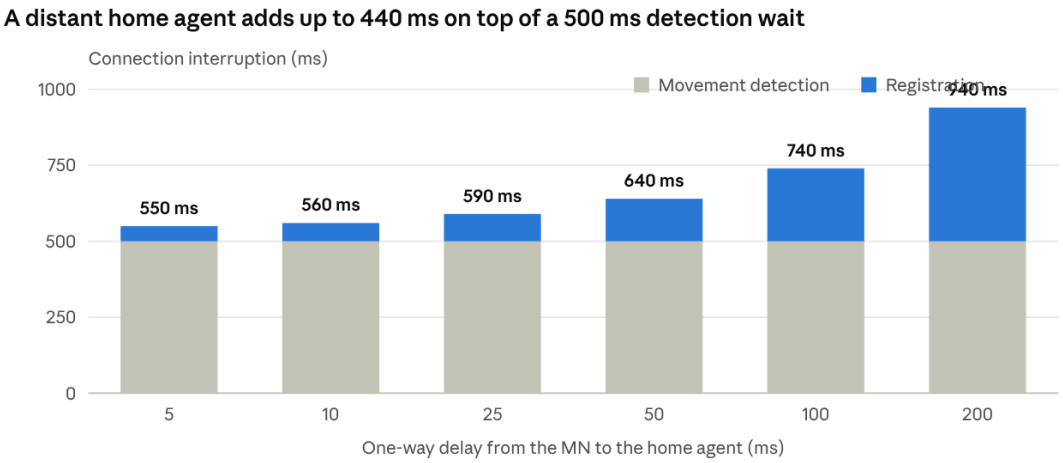
Challenge	Cause	Effect	Remedy
Tunnelling overhead	20-byte extra header on each packet	Lower efficiency, possible fragmentation	Header compression, route optimisation
Triangular routing	All CN traffic goes through the HA	Longer path, more delay, HA load	Route optimisation (MIPv6), reverse path planning
Packet loss	Packets sent to old CoA while MN is moving	Loss burst, TCP timeouts	Buffering and forwarding by old FA, smooth handoff
Security	Open wireless medium, redirection attacks	Hijacked or redirected traffic	Authentication, replay protection, encryption (IPsec)
Scalability	One binding and signalling per MN movement	HA overload, signalling storm	Hierarchical Mobile IP, distributing HAs
Ingress filtering	Source address is topologically wrong in foreign network	Packets dropped by firewalls	Reverse tunnelling

2.3.2 Registration delay

The interruption time is the sum of the wait for an advertisement and the registration round trip:

T_interrupt = T_detect + 2 x d_MN-HA + T_processing

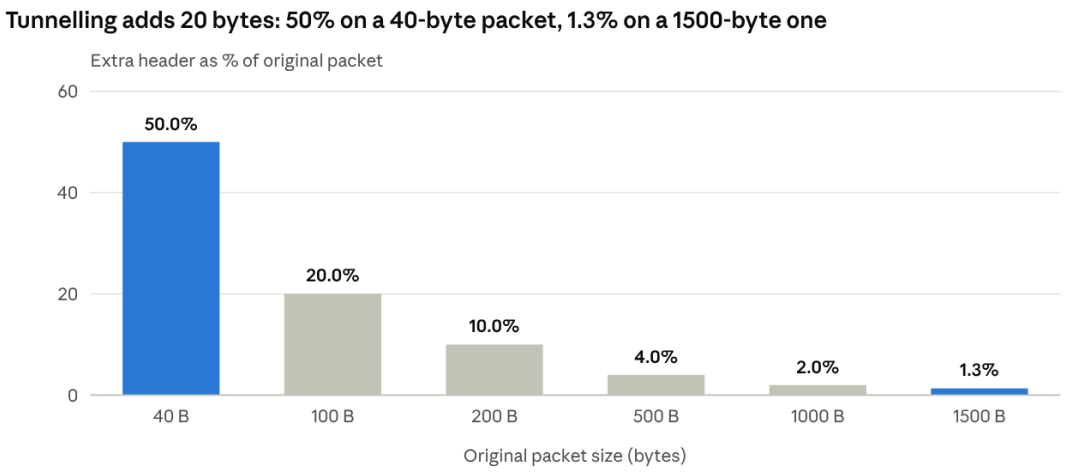
With a 1 s advertisement interval, T_detect is on average 500 ms. Here d_MN-HA is the one-way delay from the MN to the home agent (radio hop included) and the processing time is 40 ms, so the registration delay is 40 + 2 d (ms); in the worked example d = 10 + 50 = 60 ms gives 160 ms. Graph 2.2 shows how the interruption grows as the home agent gets further away: detection is a fixed 500 ms, while registration grows from 50 ms to 440 ms. At 50 packets/s, an interruption of 940 ms means about 47 lost packets per movement.



Graph 2.2 · analytical model, 1 s advertisement interval

2.3.3 Tunnelling overhead

IP-in-IP adds 20 bytes to every tunnelled packet. The relative overhead is 20 / packet size. For a 160-byte voice packet it is 12.5%, for a 40-byte TCP acknowledgement 50% and for a full-size 1500-byte packet 1.3%. If the original packet is already as large as the MTU the packet must be fragmented, doubling the number of packets.



Graph 2.1 · analytical model, 20-byte IP-in-IP header

2.3.4 Triangular routing

The CN sends to the home address, so the packet goes CN to HA to MN, while the reply goes directly MN to CN. With CN-HA 60 ms, HA-MN 70 ms and the direct CN-MN path 40 ms, the forward path takes 130 ms, which is 3.25 times the direct delay. The effect is worst when the MN and the CN are close to each other but far from the home network (for example both in the same city while the HA is on another continent).

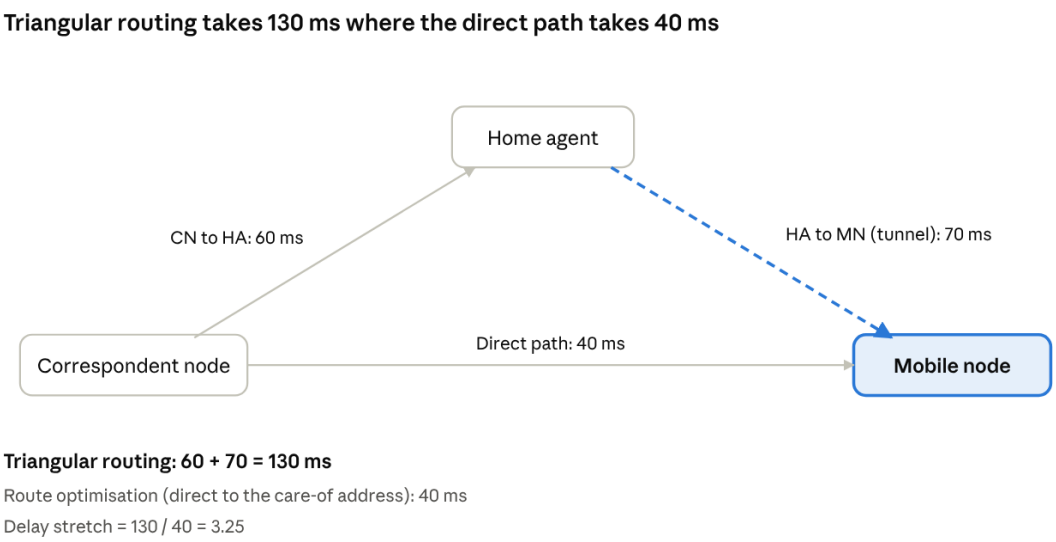


Figure 2.3 · triangular routing versus direct path

Figure 2.3 – Triangular routing versus the direct path (delays assumed for the example).

2.3.5 Packet loss during mobility

Between leaving the old network and completing registration the HA continues to tunnel packets to the old CoA, where nobody receives them. Packets lost per movement = packet rate x T_interrupt. For a 50 packets/s stream and a 940 ms interruption this is 47 packets; for TCP the loss triggers retransmission timeouts and congestion-window reduction, so the throughput drops for several seconds after each handoff. Remedies: the old FA buffers and forwards packets to the new CoA (smooth handoff), and the new FA pre-registers the MN (fast handoff).

2.3.6 Security and scalability considerations

Security. A false registration could redirect all of a user's traffic to an attacker. Protection: mandatory MN-HA authentication with a shared secret (keyed MD5/HMAC), replay protection with the identification field (timestamps or nonces), optional FA-HA authentication, and IPsec

for confidentiality. **Scalability.** The HA must store a binding for each away MN and process registrations from all of them; if many users move often, the HA and the signalling traffic are a bottleneck and a single point of failure. Hierarchical registration, several HAs for load sharing and longer lifetimes reduce the load.

```
ALGORITHM DetectMobileIPChallenge(MN, Route, Registration)
INPUT:
  MN, current route, registration state
OUTPUT:
  Identified challenge
BEGIN
  IF RegistrationStatus = FAILED THEN
    RETURN REGISTRATION_PROBLEM
  ELSE IF HandoffInProgress = TRUE THEN
    RETURN HANDOFF_DELAY
  ELSE IF PacketLossRate > MAX_LOSS THEN
    RETURN PACKET_LOSS
  ELSE IF RouteDelay > MAX_DELAY THEN
    RETURN ROUTING_DELAY
  ELSE IF SecurityValidation = FAILED THEN
    RETURN SECURITY_PROBLEM
  ELSE
    RETURN NO_MAJOR_PROBLEM
```

END IF

END

2.4 Joint Design Question

Question: If a mobile user frequently changes its network of attachment during an active communication session, what problems can occur in Mobile IP and how can they be reduced?

Problems. (1) Each change triggers a full registration with the distant HA, so interruptions of 500 to 1000 ms repeat many times per session. (2) Packets sent to the old CoA are lost during every interruption. (3) The signalling load on the HA and the MN's battery use grow with the number of changes. (4) TCP interprets the repeated losses as congestion and keeps its window small. (5) Triangular routing and tunnelling add delay and overhead to every packet. (6) Rapid changes at a cell border (ping-pong) can cause registration storms and out-of-order packets. (7) Each registration is a security risk if not authenticated.

Improvements and justification.

Technique	What it does	Effect on the problem
Regional (hierarchical) registration	Register locally with a gateway foreign agent; HA is contacted only when the region changes	Registration round trip falls from FA-HA delay to a local delay (5 ms)
Link-layer triggers / solicitation	Layer 2 tells Layer 3 immediately that attachment changed, so the MN	Detection falls from about 500 ms to about 50 ms

Technique	What it does	Effect on the problem
	solicits at once	
Fast handover with pre-registration	New FA is prepared before the move; old FA forwards buffered packets	Interruption becomes about 20 ms, loss about 1 packet
Route optimisation (MIPv6 binding update to CN)	CN sends directly to the CoA	Removes triangular routing and tunnel overhead
Hysteresis and handoff damping	Avoids ping-pong at cell borders	Fewer registrations and loss bursts
Reverse tunnel and IPsec	Handles ingress filtering and protects signalling	Keeps connections working and secure

Table 2.1 – Estimated interruption per movement (HA 200 ms away, 50 packets/s)

Scheme	Detection (ms)	Registration (ms)	Total (ms)	Packets lost per movement
Basic Mobile IP	500	440	940	47
Regional registration	500	50	550	28
Regional registration + link-layer triggers	50	50	100	5

Scheme	Detection (ms)	Registration (ms)	Total (ms)	Packets lost per movement
Fast handover with pre-registration	0	20	20	1

The best design for a frequently moving user combines local (regional) registration, layer-2 triggers and buffering at the old agent, with route optimisation for long sessions: it cuts the interruption from 940 ms to about 100 ms or less and removes the permanent dependence on the distant HA.

2.5 Evidence expected

Evidence	Where
Mobile IP architecture diagram	Figure 2.1
Registration and packet-flow sequence	Figure 2.2
Mobility-related delays and overhead	Graph 2.1, Graph 2.2, Figure 2.3, Table 2.1
Justification of the improvement	Section 2.4

2.6 Cross-understanding

- *Student A on the challenges:* the main challenges are registration delay (detection + round trip to the HA), tunnelling overhead (20 bytes per packet), triangular routing, packet loss during the interruption, security of registration and HA scalability, each with a remedy given in Section 2.3.

- *Student B on registration and delivery:* the MN hears an advertisement, obtains a CoA, sends a registration request through the FA to the HA, the HA creates the binding and replies; later packets for the home address are intercepted by the HA, encapsulated, tunnelled to the CoA, decapsulated by the FA and delivered to the MN.

Q3 – Cellular Networks versus Ad Hoc Wireless Networks (7 Marks)

3.1 Shared Group Work

3.1.1 Cellular wireless network architecture

A cellular network divides a service area into **cells**, each served by a fixed **base station (BS)** with its own set of radio channels. Mobile stations never talk to each other directly: they talk to the base station, which is connected by wired backhaul to a **base station controller (BSC)** and a **mobile switching centre (MSC)**. The MSC connects calls to the PSTN and the Internet and uses two databases, the **home location register (HLR)** and the **visitor location register (VLR)**, to know where each subscriber is.

The same frequencies are **reused** in cells that are far enough apart; neighbouring cells use different channels to avoid co-channel interference. If the total number of duplex channels is S and the cluster size is N, each cell gets S / N channels. Example: S = 56 channels and N = 7 give 8 channels per cell; a network of 100 cells then carries 800 simultaneous calls although only 56 frequencies exist.

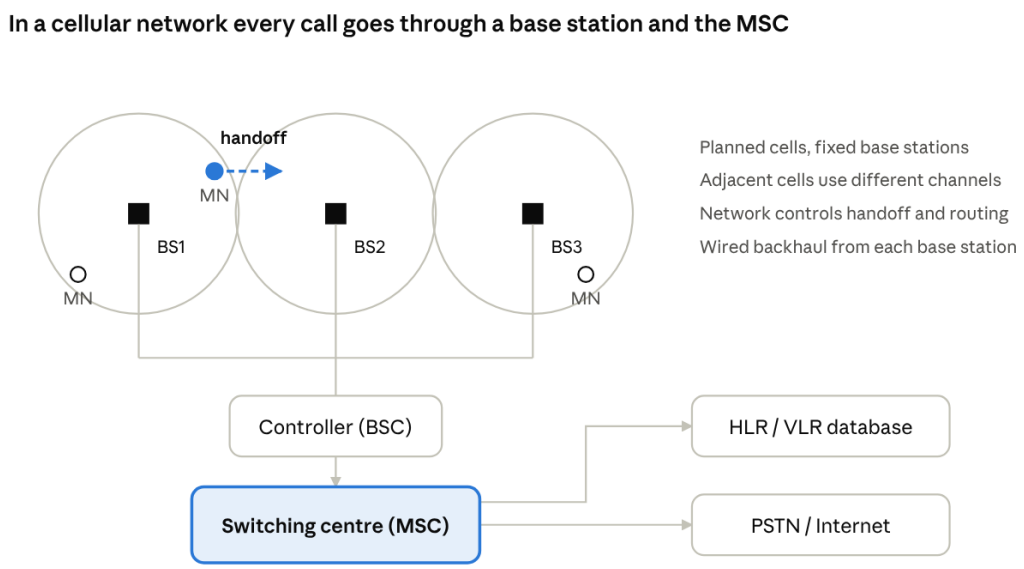


Figure 3.1 · cellular network architecture

Figure 3.1 – Cellular architecture with three cells, the controller, the MSC and the core network. The accent arrow shows a mobile node crossing a cell boundary (handoff).

3.1.2 Ad hoc wireless network architecture

An ad hoc network (MANET) is a collection of mobile nodes that form a network on their own, without base stations, access points or any fixed administration. Every node is **both a host and a router**: it sends and receives its own traffic and also forwards packets of other nodes. If two nodes are not within radio range, packets travel **multi-hop** through intermediate nodes. Because nodes move, switch off or lose links, the **topology changes dynamically**, so routes must be discovered and repaired continuously by distributed routing protocols such as DSDV, AODV and DSR.

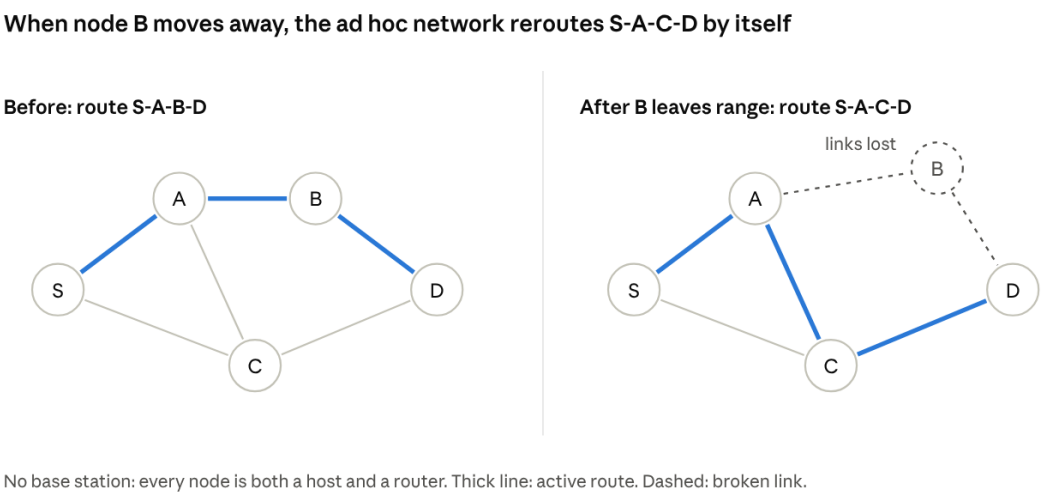


Figure 3.2 · ad hoc network before and after a topology change

Figure 3.2 – Ad hoc network before and after node B moves out of range; the route is rebuilt through C without any infrastructure.

3.1.3 Infrastructure-based and infrastructure-less communication

In **infrastructure-based** communication, all traffic is relayed by fixed devices (BS, MSC) owned and planned by an operator; the mobile node needs only one wireless hop. In **infrastructure-less** communication, the nodes themselves provide connectivity, so a path may need several wireless hops and no operator is required.

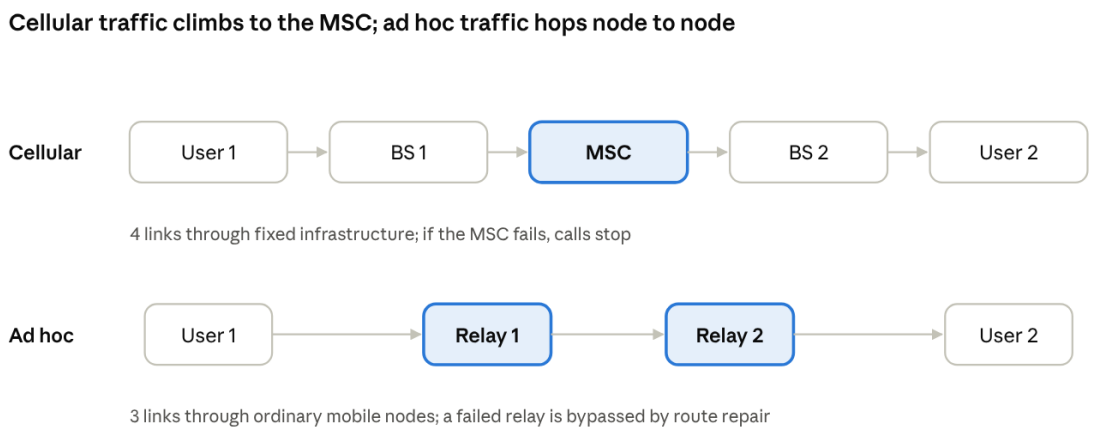


Figure 3.3 · technical comparison of the two communication paths

Figure 3.3 – Technical comparison diagram: the cellular path goes through the base stations and the MSC, the ad hoc path through relay nodes.

3.1.4 Mobility, topology and routing requirements

- **Mobility:** in cellular networks the mobile node moves, but the base stations and the topology of the infrastructure are fixed; mobility is managed by handoff and location registers. In ad hoc networks, all nodes may move, including the routers.
- **Topology:** cellular topology is a planned, static hierarchy (star of cells connected in a tree); ad hoc topology is a dynamic, arbitrary graph G = (V, E) whose edges appear and disappear with node movement.
- **Routing:** cellular routing is infrastructure-assisted and mostly static in the core; ad hoc routing is distributed, multi-hop and must react to topology changes (proactive or reactive protocols).

3.2 Student A – Cellular Network Investigation

3.2.1 Cellular architecture representation

Represented in Figure 3.1: mobile station, BS, BSC, MSC, HLR/VLR and PSTN/Internet gateways form a hierarchy; each BS covers one cell of radius 0.5 to 5 km (macro cell) or less (micro and pico cells).

3.2.2 Role of base stations and infrastructure

The base station provides the radio interface (transmit, receive, power control, channel allocation), measures signal quality of its mobiles, and forwards traffic to the BSC. The BSC manages many BSs, allocates radio channels and decides on handoffs between its cells. The MSC switches calls, interconnects with other networks, performs authentication and billing with the HLR/VLR and manages inter-MSC handoff. This fixed infrastructure gives guaranteed coverage and quality, but also means that service exists only where the operator has built it.

3.2.3 Centralised communication

A call from User 1 to User 2 follows User 1, BS1, BSC, MSC, BSC, BS2, User 2 (Figure 3.3). Control (call setup, authentication, paging, handoff decisions) is centralised in the infrastructure; the mobile only has to measure and obey. This simplifies the terminals and gives the operator tight control of quality and security, at the cost of dependency on the infrastructure and a minimum delay of about 40 ms even for two nearby users.

3.2.4 Handoff between cells

A handoff keeps a call alive when the mobile leaves a cell. Steps: (1) the MN or the BS measures signal strength; (2) if the serving signal falls below a threshold and a neighbour is stronger by a

margin, the BSC decides to hand off; (3) the target BS reserves a channel; (4) the MN is told to retune; (5) the MSC switches the path to the new BS and releases the old channel. **Hard handoff** (break-before-make, GSM) causes a short interruption of 50 to 100 ms; **soft handoff** (make-before-break, CDMA) keeps links to two cells simultaneously and has no interruption. If no channel is free in the target cell, the call is **dropped** (handoff failure); operators therefore reserve guard channels for handoffs because dropping a call is worse than blocking a new one.

3.2.5 Scalability and coverage

Capacity scales by adding cells, not by adding spectrum: **cell splitting** (halving the radius multiplies the number of cells and the capacity by about 4), **sectorisation** (3 sectors per cell reduce interference and raise reuse) and **frequency reuse**. Example: a 100-cell network with 8 channels per cell carries 800 calls; splitting every cell into 4 gives 400 cells and 3200 calls with the same spectrum. Coverage is limited by base-station placement, so rural and disaster areas are weakly covered.

3.2.6 Suitable cellular applications

Public voice and SMS, mobile broadband (4G/5G), video streaming, mobile banking and payments, IoT over NB-IoT/LTE-M, vehicle telematics and any service needing wide-area coverage, guaranteed quality and billing.

3.3 Student B – Ad Hoc Network Investigation

3.3.1 Ad hoc architecture representation

Represented in Figure 3.2 as a graph G = (V, E): V is the set of mobile nodes, E the set of currently usable links (distance below the transmission range R and signal above MIN_SIGNAL). There is no distinguished node; the network is formed by neighbour discovery with HELLO messages.

3.3.2 Communication between mobile nodes

Nodes communicate directly when within range and otherwise through relays. Medium access is by CSMA/CA (IEEE 802.11 ad hoc mode) with the hidden-terminal and exposed-terminal problems; every node forwards packets according to its routing table or a discovered route. Peer-to-peer services (file exchange, chat, sensor reporting) need no server.

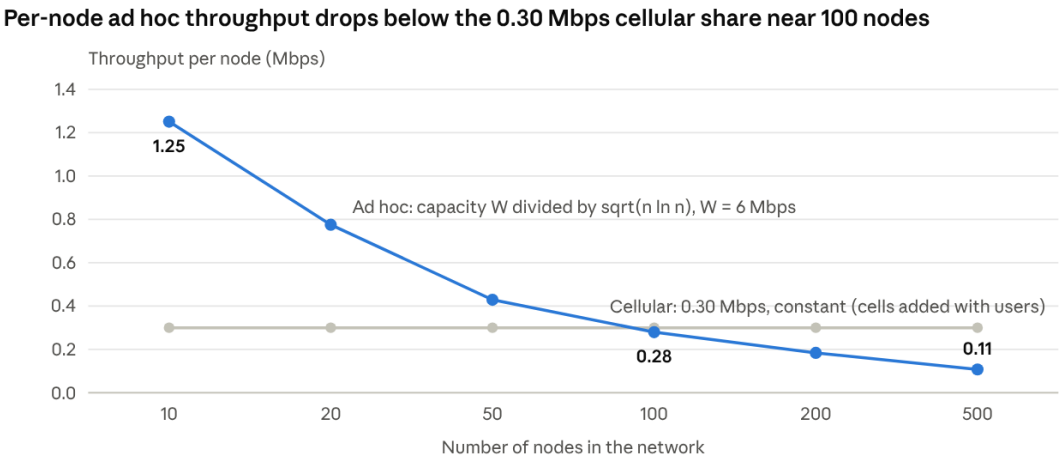
3.3.3 Dynamic topology

Link lifetime is roughly R / v where R is the radio range and v the relative speed. For R = 100 m, a pedestrian pair (v = 1.5 m/s) keeps a link about 67 s, vehicles at 15 m/s only about 7 s, and vehicles in opposite directions (relative speed 30 m/s) only about 3 s. Each broken link may break all routes that use it, so route maintenance overhead grows with mobility.

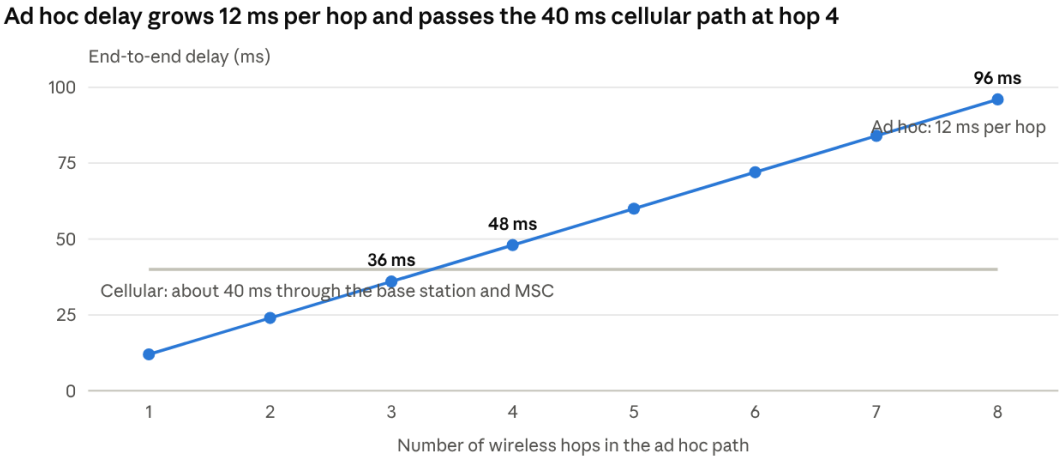
3.3.4 Multi-hop communication

Multi-hop extends coverage without infrastructure, but each hop adds delay and consumes the shared channel. With 12 ms per hop (contention, transmission, processing) the end-to-end delay is 12 x hops ms, which exceeds the 40 ms of a cellular path from the fourth hop on (Graph 3.2). The per-node capacity also falls as the network grows, since each packet is transmitted by

several nodes: the theoretical scaling is W / sqrt(n ln n) (Gupta and Kumar). Graph 3.1 compares it with a cellular user, whose share stays constant at 6 Mbps / 20 users = 0.30 Mbps because new cells are added as users grow.



Graph 3.1 · capacity-scaling model, W = 6 Mbps, 20 users per cell



Graph 3.2 · 12 ms per wireless hop, 40 ms cellular path

Graph 3.1 and Graph 3.2 – Scaling of throughput and delay: cellular networks buy scalability with infrastructure; ad hoc networks lose per-node throughput and delay as nodes and hops increase.

3.3.5 Absence of fixed infrastructure

Ad hoc networks need no towers, wiring, planning or subscription; they can be deployed in minutes and survive when a node fails because other paths may exist. The price: no central control for security, addressing, QoS or billing; limited battery and bandwidth; and difficult management.

3.3.6 Suitable ad hoc applications

Disaster recovery and rescue communication, military and battlefield networks, temporary events and conferences, vehicular networks (VANET), wireless sensor networks, community mesh networks and device-to-device sharing.

3.4 Comparative Analysis

Table 3.1 – Cellular network versus ad hoc network

Parameter	Cellular Network	Ad Hoc Network
Infrastructure	Fixed base stations, controllers, MSC and core network, owned and planned by an operator	None; nodes are hosts and routers and form the network by themselves
Communication	Single wireless hop to the BS, then	Peer-to-peer over one or many

Parameter	Cellular Network	Ad Hoc Network
	wired infrastructure; centralised	wireless hops; distributed
Routing	Infrastructure-assisted, static, handled by core network	Distributed multi-hop routing (DSDV, AODV, DSR) which must adapt to changes
Mobility	Mobile node moves, infrastructure fixed; managed by handoff and HLR/VLR	All nodes may move, including routers; handled by route maintenance
Topology	Planned hierarchical cells (star/tree)	Dynamic arbitrary graph with changing links
Scalability	Scales by adding cells, frequency reuse and cell splitting; per-user rate stays constant	Per-node throughput falls about 1 / sqrt(n ln n); control overhead grows with nodes and mobility
Handoff	Network-controlled hard or soft handoff between cells	No handoff; route repair replaces the broken hop
Failure handling	Failure of a BS or MSC removes service for an area (backup links and overlapping cells reduce impact)	A failed node is bypassed if an alternative path exists; network degrades gracefully
Applications	Public voice, mobile broadband,	Disaster recovery, military,

Parameter	Cellular Network	Ad Hoc Network
	IoT, telematics	VANET, sensor and temporary networks

3.5 Joint Engineering Decision

Question: For an emergency or disaster environment where fixed infrastructure may be unavailable, which network architecture would you select and why?

Decision: an ad hoc (multi-hop) wireless network, with gateway nodes linked to any surviving infrastructure or satellite as an optional uplink.

```
ALGORITHM SelectWirelessArchitecture(Scenario)
INPUT:
  InfrastructureAvailability, NodeMobility, DeploymentTime,
  CoverageRequirement, MultiHopRequirement
OUTPUT:
  Recommended architecture
BEGIN
  IF InfrastructureAvailability = AVAILABLE AND
     CoverageRequirement = LARGE THEN
    Architecture <- CELLULAR
  ELSE IF DeploymentTime = IMMEDIATE AND
          InfrastructureAvailability = UNAVAILABLE THEN
    Architecture <- AD_HOC
```

```
  ELSE IF MultiHopRequirement = REQUIRED THEN
    Architecture <- AD_HOC
  ELSE
    Architecture <- CELLULAR
  END IF
  RETURN Architecture
END
```

For the disaster scenario: InfrastructureAvailability = UNAVAILABLE, DeploymentTime = IMMEDIATE, MultiHopRequirement = REQUIRED, so the algorithm returns AD_HOC.

Reasons. (1) Towers, power and backhaul are often damaged or overloaded in a disaster, and a cellular network cannot work without them. (2) Ad hoc nodes can be deployed immediately by rescue teams, without planning. (3) Multi-hop forwarding extends coverage over rubble and difficult terrain. (4) The network has no single point of failure, so it degrades gracefully when nodes are lost. (5) Rescue teams mainly communicate locally, which suits peer-to-peer traffic. **Limits:** lower capacity per node, higher delay with many hops, battery constraints and weaker security; they are reduced by choosing a suitable routing protocol, keeping paths short and adding gateway nodes.

3.6 Cross-Defence

- *Student A – why cellular networks depend on infrastructure:* the mobile has only one radio hop, so it needs a base station to reach anything; call routing, authentication, channel allocation and handoff are performed by the BS, BSC, MSC and registers. If they fail or lose power, coverage disappears in the whole area, because there is no mechanism by which handsets forward each other's traffic.

- *Student B – how ad hoc networks continue without fixed infrastructure:* each node discovers its neighbours by HELLO messages, keeps or discovers routes with a distributed routing protocol, forwards packets of others and, if a link or node fails, repairs the route through another neighbour (Figure 3.2).

Q4 – Applications of Mobile IP and Routing Requirements (7 Marks)

4.1 Shared Group Work

4.1.1 Real-world applications requiring mobility support

Application	Mobile entities	Why mobility support is needed
Emergency communication	Rescue teams, police and fire vehicles	Sessions must continue while units move between areas
Mobile healthcare	Ambulances, patient monitors, doctors with tablets	Life-critical data must stay connected to the hospital
Intelligent transportation	Vehicles, buses, roadside units	Continuous traffic, safety and infotainment services across networks
Mobile workforce	Employees with laptops and phones	Access to company services while moving between Wi-Fi and cellular

Application	Mobile entities	Why mobility support is needed
Fleet and logistics	Trucks, containers, delivery vehicles	Tracking and control by a fixed head office
Field engineering	Maintenance crews, remote equipment	Reach to internal systems from changing locations
Temporary networks	Event and construction-site devices	Connectivity without fixed addressing

4.1.2 Selected application scenario: ambulance tele-medicine (mobile healthcare)

The group selected **mobile healthcare**: an ambulance carries a patient to the hospital while the paramedic's gateway streams vital signs (ECG, SpO2, blood pressure), voice and live video to a doctor at the hospital, who guides the treatment. It was selected because the session is life-critical, long-lasting (10 to 30 minutes), crosses many cells and several operator networks, and cannot be restarted after each handoff.

4.1.3 Mobile entities and communication requirements

Flow	Rate and packet size	One-way delay target	Loss target
Vital-sign telemetry	10 packets/s, 200 bytes (about 16 kbps)	150 ms	1% or less
Voice with the doctor	50 packets/s, 160 bytes (about 64 kbps)	150 ms	2% or less

Flow	Rate and packet size	One-way delay target	Loss target
Live video	about 500 kbps	300 ms	3% or less
Alerts and GPS location	event driven, small	100 ms	reliable delivery

Mobile entities: the ambulance mobile router (the Mobile Node for Mobile IP) with its patient sensors attached over a personal-area network, and the paramedic's tablet. **Fixed endpoints:** the hospital telemedicine server and electronic health record (the Correspondent Node) and the doctor's console. **Network entities:** the hospital's Home Agent, and Foreign Agents of the cellular or Wi-Fi networks along the route.

4.1.4 Applying Mobile IP to the scenario

The ambulance gateway is given a permanent home address from the hospital network (for example 10.20.0.50) and the hospital's router acts as its Home Agent. Every application session (telemetry, voice, video) is bound to this address, so it does not change when the ambulance switches from cell A to cell B or from one operator to another. On each change the gateway obtains a care-of address, registers with the Home Agent and the HA tunnels packets to the new CoA. Because the Correspondent Node (the hospital server) is in the home network, the HA is directly on the path and **triangular routing adds almost no extra delay**, which is a key design choice. A reverse tunnel is used because the operators' ingress filters would otherwise drop packets whose source address (the hospital address) does not belong to the visited network, and IPsec protects the patient data.

4.1.5 Routing requirements created by mobility

- **Reachability of the current location:** the network must always know the current CoA (binding freshness), so registration must be fast.
- **Fast route repair:** when a link or cell changes, the new path must be installed in tens of milliseconds, not seconds.
- **Short paths:** routes through the HA should not add large detours; the HA is placed in the hospital network, near the CN.
- **Low overhead:** signalling must not use bandwidth needed by vitals and video.
- **Loop-free and stable routing,** even when the ambulance moves at 100 km/h and cells change every minute.
- **Alternative routes:** if no infrastructure is available (a tunnel, a disaster), the gateway should be able to relay through other vehicles using ad hoc routing (AODV or DSDV).
- **Security of routing and registration messages** so that nobody can redirect the stream.

4.1.6 Evaluation of the proposed solution

Mobile IP satisfies the main need (session continuity with an unchanged address), but the standard procedure has a long interruption at every handoff. The solution is suitable if combined with regional registration, link-layer triggers and buffering (Section 4.3); then the loss and delay targets are met.

4.2 Student B – Application Architecture

4.2.1 Application selected and description

Ambulance tele-medicine. A paramedic attaches the sensors to the patient; the ambulance gateway collects the data, forms IP packets and sends them to the hospital server. The doctor sees the data in real time and speaks to the paramedic.

4.2.2 Mobile nodes and communication endpoints

Entity	Role	Mobility
Ambulance mobile router	Mobile Node of Mobile IP; connects the in-vehicle network to the outside	Moves at up to 100 km/h
Patient sensors, tablet	Hosts behind the mobile router (network mobility)	Move with the ambulance
Foreign Agents (cell A, cell B, ...)	Provide CoAs and decapsulate packets	Fixed
Home Agent	Keeps the binding, tunnels packets	Fixed, in hospital
Telemedicine server / EHR	Correspondent Node	Fixed
Doctor console	User endpoint	Fixed or mobile

4.2.3 Application network architecture

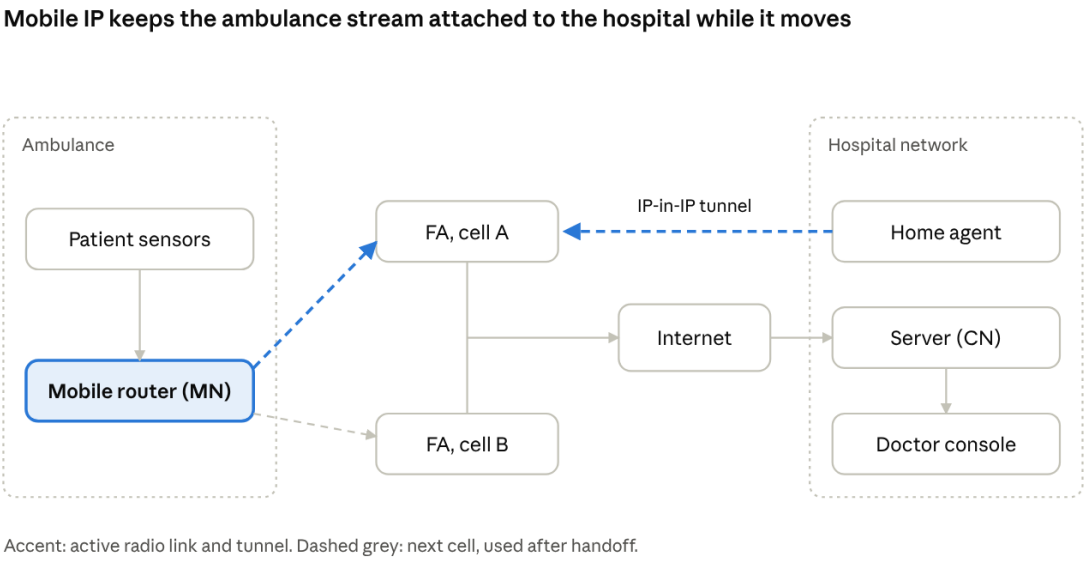


Figure 4.1 · ambulance telemedicine architecture with Mobile IP

Figure 4.1 – The sensors reach the hospital through the mobile router, the Foreign Agent of the current cell, the Internet and the Home Agent in the hospital network.

4.2.4 How Mobile IP supports user mobility

While the ambulance is at the hospital, its home address 10.20.0.50 is reachable directly. After leaving, the gateway attaches to cell A and receives CoA A; it registers CoA A with the HA, and the HA now intercepts packets for 10.20.0.50 and tunnels them to CoA A. When it crosses into cell B, it obtains CoA B and sends a new registration; the HA replaces the binding, and the telemetry and voice sessions, which only know 10.20.0.50, continue without being reopened. TCP sessions (patient record upload) do not break because their endpoint addresses never change.

4.2.5 Required network components

Component	Function
Mobile router with two radios (4G/5G and Wi-Fi)	Maintains the attachment and allows make-before-break handoff
Home Agent in hospital	Binding table, tunnelling, authentication of the gateway
Foreign Agents or GFA	Care-of addresses and local (regional) registration
Hospital server and EHR	Receives and stores telemetry
AAA server (RADIUS)	Authenticates the gateway and the user
IPsec gateway	Confidentiality of patient data
Monitoring and buffer module in mobile router	Stores vitals locally if the link is lost

4.2.6 Communication flow

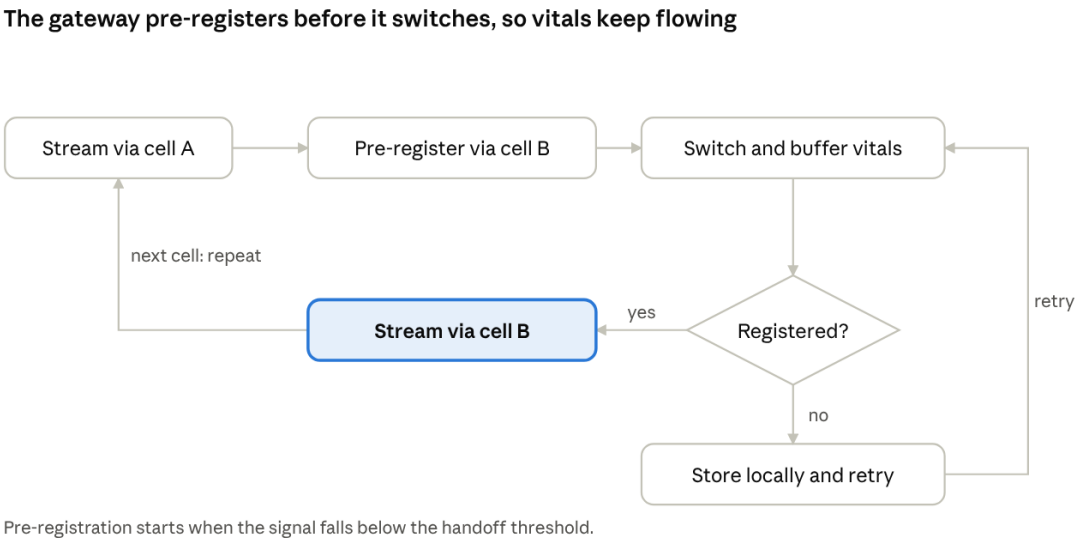


Figure 4.2 · communication flow during a handoff

Figure 4.2 – The flow during a handoff: pre-registration, switching with buffering, registration check, and local storage with retry on failure.

ALGORITHM PerformHandoff(MN, OldNetwork, NewNetwork)

INPUT:

MN - mobile node (ambulance gateway)

OldNetwork - previous point of attachment

NewNetwork - target network

OUTPUT:

Handoff status

BEGIN

Detect NewNetwork

Measure signal quality

```
  IF SignalQuality(NewNetwork) < MIN_SIGNAL THEN
    RETURN HANDOFF_REJECTED
  END IF
  Obtain care-of address for NewNetwork
  Pause non-critical transmissions (video)
  UPDATE mobility registration
  UPDATE routing information
  IF RegistrationStatus = SUCCESS THEN
    SWITCH active attachment TO NewNetwork
    Resume data transmission and flush buffer
    RETURN HANDOFF_SUCCESS
  ELSE
    Restore OldNetwork if available
    RETURN HANDOFF_FAILURE
  END IF
END
```

4.3 Student A – Performance and Requirement Analysis

4.3.1 Communication delay

Table 4.1 – One-way delay budget of the telemetry flow (MN to doctor console)

Component	Delay (ms)

Component	Delay (ms)
Sensor sampling and packetisation	20
Radio uplink (shared channel, retransmissions)	25
Operator core network	15
Internet path to hospital	20
Tunnel handling (decapsulation, reverse tunnel)	5
Server processing and display	10
Total	**95**

The total of 95 ms is below the 150 ms target. Because the CN is in the home network, the HA adds only about 5 ms; if the HA were far from the hospital (triangular routing, Figure 2.3), the delay would grow to 130 ms or more and the margin would disappear.

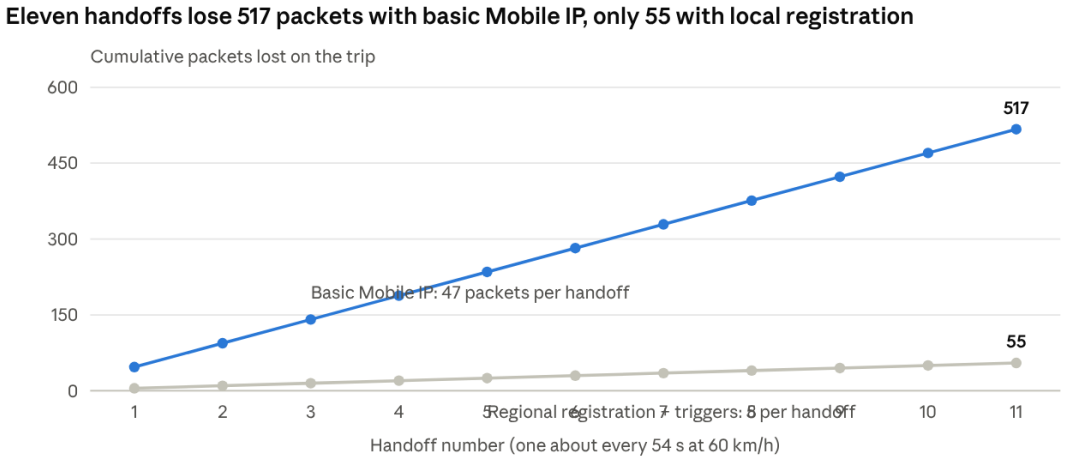
4.3.2 Packet delivery and connectivity

The ambulance travels for 10 minutes (600 s) at 60 km/h through cells 900 m apart, so a handoff occurs about every 54 s: **11 handoffs**. At 50 packets/s a total of 30 000 packets are sent. The worst-case HA distance (200 ms one-way) gives the interruption values of Table 2.1.

Table 4.2 – Delivery during the trip

Scheme	Interruption per handoff (ms)	Packets lost per handoff	Packets lost on trip	Delivery ratio	Meets 1% telemetry target

Scheme	Interruption per handoff (ms)	Packets lost per handoff	Packets lost on trip	Delivery ratio	Meets 1% telemetry target
Basic Mobile IP	940	47	517	98.28%	No
Regional registration	550	28	308	98.97%	Yes (just)
Regional registration + triggers	100	5	55	99.82%	Yes
Fast handover, pre-registration	20	1	11	99.96%	Yes



Graph 4.1 · 11 handoffs in a 10-minute trip, 50 packets/s, HA 200 ms away

Basic Mobile IP meets the 2% voice target but not the 1% telemetry target, and the losses arrive as bursts of 47 consecutive packets (almost 1 s without vital signs), which is more serious than the average suggests. Connectivity gaps also occur in coverage holes (tunnels, valleys); the store-and-forward buffer in the gateway keeps the data until the link returns.

4.3.3 Mobility requirements

- Session continuity at speeds up to 100 km/h (27.8 m/s, a cell crossing about every 32 s).
- Fixed identity: the sessions must keep the home address 10.20.0.50 across operators.
- Location update within 100 ms of the attachment change.
- Support for two simultaneous attachments (make-before-break) to avoid gaps.
- Support for network mobility: the sensors and tablet behind the router must not register individually.

4.3.4 Interruption during movement

The interruption is T = T_detect + registration. In basic Mobile IP with an advertisement every 1 s and the HA 200 ms away it is 500 + 440 = 940 ms. Even for a typical HA 30 ms away it is 500 + 100 = 600 ms. For a vitals stream this is a gap of up to one second at each of the 11 handoffs; for voice it is audible as a drop-out. With regional registration and link-layer triggers the gap is reduced to about 100 ms, which the receiver's jitter buffer (150 ms) can hide.

4.3.5 Security and reliability requirements

Requirement	Mechanism
Authentication of the gateway to the HA	Shared secret or certificate; keyed hash in registration request
Protection against replay and redirection of the stream	Identification field (timestamp or nonce), lifetime limits
Confidentiality of patient data	IPsec ESP tunnel between mobile router and

Requirement	Mechanism
	hospital
Integrity of control messages	Authentication extension on advertisements and registrations
Availability when the HA fails	Redundant HAs (VRRP/HSRP), buffered data
Reliability across coverage holes	Dual radio interfaces, store-and-forward, retransmission of critical alerts
Auditability	Logging of registrations and accesses for clinical records

Conclusion: basic Mobile IP is only partly suitable (64%) because of the long interruptions, but Mobile IP with regional registration, link-layer triggers, buffering, reverse tunnelling and IPsec is suitable (82%) for ambulance tele-medicine.

4.4 Joint Application Evaluation

Aspect	Result
Application	Ambulance tele-medicine with a mobile router and a hospital home agent
Why Mobile IP	Keeps the address and the sessions unchanged across cells and operators

Aspect	Result
Main routing requirements	Fresh bindings, fast repair, short paths, low overhead, alternative routes
Delay	95 ms against the 150 ms target
Packet delivery	99.82% with enhancements, 98.28% with basic Mobile IP
Interruption	100 ms (enhanced) versus 940 ms (basic) per handoff
Security	Authentication, replay protection and IPsec
Suitability	Suitable with the enhancements listed in Section 4.3